

Based on the code provided for your safety system, here are the complete wiring connections. This setup requires an Arduino Board (like an Uno), an MQ series gas sensor, an HW-028 (or similar) rain sensor, two LEDs (with resistors), and a buzzer.

Step-by-Step Connection Guide

This project is best assembled using a common power and ground rail on a breadboard.

Important Voltage Note: Most MQ gas sensors contain a small internal heater that requires 5V to function properly. **Do not attempt to power this circuit from the Arduino's 3.3V pin.**

Common Rails

1. Connect **Arduino 5V** to your breadboard's positive (+) rail.
2. Connect **Arduino GND** to your breadboard's negative (-) rail.

Component Connections

1. Rain Sensor (HW-028 / HW-484 or similar)

These sensors typically consist of a flat sensing board and a separate, small controller module with four pins.

1. **VCC (Controller Module):** Connect to the **5V rail**.
2. **GND (Controller Module):** Connect to the **GND rail**.
3. **DO (Digital Out):** Connect to Arduino Pin **D2** (`const int rainSensorPin = 2;`).
4. *(AO (Analog Out): Not used in this specific code).*
5. *Also, connect the two separate pins on the sensor plate back to the two input terminals on the controller module using the supplied wire.*

2. Gas/Smoke Sensor (MQ-2, MQ-5, etc.)

These modules usually have four pins.

1. **VCC:** Connect to the **5V rail**.
2. **GND:** Connect to the **GND rail**.
3. **AO (Analog Out):** Connect to Arduino Pin **A2** (`const int gasSensorPin = A2;`).
4. *(DO (Digital Out): Not used in this code).*

3. Buzzer

This can be an active buzzer or a simple piezo buzzer (if using **digitalWrite()** as the code does, an active buzzer will give a louder beep).

1. **Positive (+) terminal (Long leg):** Connect to Arduino Pin **D11** (`const int buzzerPin = 11;`).
2. **Negative (-) terminal (Short leg):** Connect to the **GND rail**.

4. Rain Alert LED (e.g., Green or Blue)

We must include a 220Ω resistor to protect the Arduino pin and the LED.

1. **Anode (Positive, long leg):** Connect to Arduino Pin **D12** (`const int ledRain = 12;`).
2. **Cathode (Negative, short leg):** Connect to one side of a **220Ω resistor**.
3. Connect the other side of the **220Ω resistor** to the **GND rail**.

5. Gas Alert LED (e.g., Red or Yellow)

We also need a 220Ω resistor for this LED.

1. **Anode (Positive, long leg):** Connect to Arduino Pin **D13** (const int ledGas = 13;).
2. **Cathode (Negative, short leg):** Connect to one side of a separate **220Ω resistor**.
3. Connect the other side of the **220Ω resistor** to the **GND rail**.